

# An Analysis of Fish Data in the North East Texas and South East Oklahoma

In this file I have combined all of the scripts, functions, and figures that I have found insightful or relevant towards the interpretation of this data.

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## Importing all of the Necessary Libraries

As I write this code, I will import **ALL** of the libraries I use in this cell

```
In [1]: import pandas as pd                # library for data
import numpy as np                        # library for vectors
import matplotlib.pyplot as plt          # library for data
import seaborn as sns                    # library for data
from sklearn.model_selection import train_test_split # library for split
from python_functions.label_plot_edges import label_plot_edges # function to label
```

## Loading the Files as Pandas Data Frames

Loading the files as pandas data frames and then cleaning them up by looping through them and removing any unwanted whitespace and characters.

| File                                            | AS | Variable |
|-------------------------------------------------|----|----------|
| gov_envData10dts_2025.csv                       | as | g_ed     |
| gov_envData10dtsMetrics_2026.csv                | as | g_edm    |
| seth_ecoRegionData_2025.csv                     | as | s_erd    |
| seth_envData_2025.csv                           | as | s_ed     |
| seth_envGenusCountData_2025.csv                 | as | s_egcd   |
| seth_fishCountData_2025.csv                     | as | s_fcd    |
| seth_genusCountData_2025.csv                    | as | s_gcd    |
| seth_gov_envData10dtsMetrics_genusCount2026.csv | as | sg_edmgc |

```
In [ ]: g_ed = pd.read_csv('../data/gov_envData10dts_2025.csv')
g_edm = pd.read_csv('../data/gov_envData10dtsMetrics_2026.csv')
s_erd = pd.read_csv('../data/seth_ecoRegionData_2025.csv')
s_ed = pd.read_csv('../data/seth_envData_2025.csv')
s_egcd = pd.read_csv('../data/seth_envGenusCountData_2025.csv')
s_fcd = pd.read_csv('../data/seth_fishCountData_2025.csv')
s_gcd = pd.read_csv('../data/seth_genusCountData_2025.csv')
sg_edmgc = pd.read_csv('../data/seth_gov_envData10dtsMetrics_genusCount2026.csv')

# ===== Put all DataFrames in a List for batch processing =====

datasets = [g_ed, g_edm, s_erd, s_ed, s_egcd, s_fcd, s_gcd, sg_edmgc]

# ===== Clean all column names =====

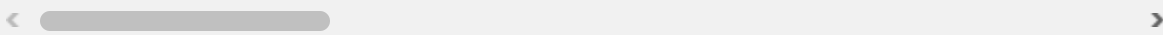
for df in datasets:
    df.columns = (
        df.columns
        .str.strip()                # remove leading/trailing spaces
        .str.replace(' ', '_')     # replace internal spaces with underscores
        .str.replace('\n', '', regex=False) # remove newlines
    )
```

```
In [3]: # g_ed.info()          # check the structure of the data frame
g_ed.head()          # printing the first 5 rows of the data frame
```

Out[3]:

|   | Segment_10Day | site_0.burn | site_0.dfm1000hr | site_0.dfm100hr | site_0.energy_ERC | site_0 |
|---|---------------|-------------|------------------|-----------------|-------------------|--------|
| 0 | 0             | 23.4        | 20.15            | 21.12           | 18.1              |        |
| 1 | 1             | 25.5        | 19.36            | 17.29           | 24.5              |        |
| 2 | 2             | 25.3        | 18.09            | 15.66           | 28.3              |        |
| 3 | 3             | 30.9        | 17.65            | 16.91           | 29.1              |        |
| 4 | 4             | 16.2        | 17.47            | 17.54           | 23.8              |        |

5 rows × 829 columns

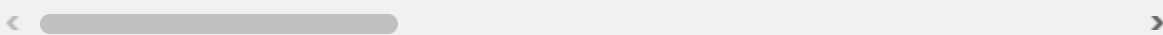


In [4]: `# g_edm.info()` *# check the structure of the data frame*  
`g_edm.head()` *# printing the first 5 rows of the data frame*

Out[4]:

|   | site   | burn_mean | burn_median | burn_sd  | burn_min | burn_max | burn_range | burn_cv  |
|---|--------|-----------|-------------|----------|----------|----------|------------|----------|
| 0 | site_1 | 24.772945 | 24.40       | 8.686572 | 5.7      | 57.2     | 51.5       | 0.350648 |
| 1 | site_2 | 24.673630 | 24.25       | 8.591869 | 4.3      | 57.3     | 53.0       | 0.348221 |
| 2 | site_3 | 25.542466 | 25.20       | 8.994322 | 3.0      | 59.1     | 56.1       | 0.352132 |
| 3 | site_4 | 26.318836 | 26.30       | 9.289094 | 2.7      | 62.3     | 59.6       | 0.352945 |
| 4 | site_5 | 24.677854 | 24.30       | 8.547560 | 5.5      | 55.3     | 49.8       | 0.346366 |

5 rows × 145 columns



In [5]: `# s_erd.info()` *# check the structure of the data frame*  
`s_erd.head()` *# printing the first 5 rows of the data frame*

Out[5]:

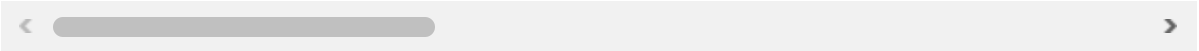
|   | Site_N | eco_Number | eco_Letter | Title                        | State |
|---|--------|------------|------------|------------------------------|-------|
| 0 | 1      | 33         | a          | Northern Post Oak Savana     | TX    |
| 1 | 2      | 33         | a          | Northern Post Oak Savana     | TX    |
| 2 | 3      | 33         | a          | Northern Post Oak Savana     | TX    |
| 3 | 4      | 32         | a          | Northern Black Land Prairie  | TX    |
| 4 | 5      | 35         | c          | Pleistocene Fluvial Terraces | TX    |

In [6]: `# s_ed.info()` *# check the structure of the data frame*  
`s_ed.head()` *# printing the first 5 rows of the data frame*

Out[6]:

|   | SiteN | State | Lat       | Long       | DO   | pH   | avg_WW   | avg_Depth | pf_Pool    | pf_    |
|---|-------|-------|-----------|------------|------|------|----------|-----------|------------|--------|
| 0 | 1     | TX    | 33.741217 | -95.927376 | 10.7 | 6.99 | 5.76666  | 23.75000  | 0.000000   | 83.333 |
| 1 | 2     | TX    | 33.717896 | -95.911323 | 4.2  | 7.22 | 2.94000  | 27.00000  | 60.000000  | 20.000 |
| 2 | 3     | TX    | 33.747466 | -96.019146 | 7.4  | 7.58 | 11.21666 | 46.33333  | 66.666667  | 33.333 |
| 3 | 4     | TX    | 33.554952 | -96.169601 | 7.7  | 7.86 | 10.51666 | 18.41666  | 0.000000   | 66.666 |
| 4 | 5     | TX    | 33.823594 | -95.861067 | 6.8  | 7.42 | 22.00000 | 39.66666  | 100.000000 | 0.000  |

5 rows × 26 columns

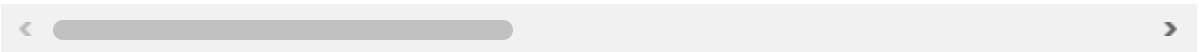


```
In [7]: # s_egcd.info()      # check the structure of the data frame
        s_egcd.head()    # printing the first 5 rows of the data frame
```

Out[7]:

|   | SiteN | State | Lat       | Long       | DO   | pH   | avg_WW   | avg_Depth | pflow_Pool | pflow  |
|---|-------|-------|-----------|------------|------|------|----------|-----------|------------|--------|
| 0 | 1     | TX    | 33.741217 | -95.927376 | 10.7 | 6.99 | 5.76666  | 23.75000  | 0.000000   | 83.333 |
| 1 | 2     | TX    | 33.717896 | -95.911323 | 4.2  | 7.22 | 2.94000  | 27.00000  | 60.000000  | 20.000 |
| 2 | 3     | TX    | 33.747466 | -96.019146 | 7.4  | 7.58 | 11.21666 | 46.33333  | 66.666667  | 33.333 |
| 3 | 4     | TX    | 33.554952 | -96.169601 | 7.7  | 7.86 | 10.51666 | 18.41666  | 0.000000   | 66.666 |
| 4 | 5     | TX    | 33.823594 | -95.861067 | 6.8  | 7.42 | 22.00000 | 39.66666  | 100.000000 | 0.000  |

5 rows × 58 columns

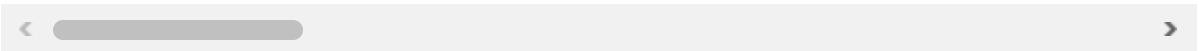


```
In [8]: # s_fcd.info()      # check the structure of the data frame
        s_fcd.head()      # printing the first 5 rows of the data frame
```

Out[8]:

|   | SiteN | State | Etheostoma_radiosum | Lepomis_megalotis | Lepomis_macrochirus | Lepomis_g... |
|---|-------|-------|---------------------|-------------------|---------------------|--------------|
| 0 | 1     | 1     |                     | 0                 | 6                   | 5            |
| 1 | 2     | 1     |                     | 0                 | 0                   | 0            |
| 2 | 3     | 1     |                     | 0                 | 24                  | 12           |
| 3 | 4     | 1     |                     | 0                 | 43                  | 54           |
| 4 | 5     | 1     |                     | 0                 | 16                  | 16           |

5 rows × 54 columns

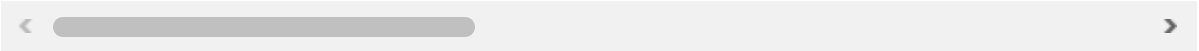


```
In [9]: # s_gcd.info()      # check the structure of the data frame
        s_gcd.head()      # printing the first 5 rows of the data frame
```

Out[9]:

|   | SiteN | State | Ameiurus | Aphredoderus | Aplodinotus | Campostoma | Cyprinella | Cyprinus |
|---|-------|-------|----------|--------------|-------------|------------|------------|----------|
| 0 | 1     | TX    | 0        | 0            | 0           | 1          | 0          | 0        |
| 1 | 2     | TX    | 0        | 0            | 0           | 0          | 0          | 0        |
| 2 | 3     | TX    | 0        | 0            | 0           | 0          | 0          | 0        |
| 3 | 4     | TX    | 0        | 0            | 0           | 36         | 39         | 1        |
| 4 | 5     | TX    | 0        | 0            | 1           | 0          | 22         | 0        |

5 rows × 32 columns

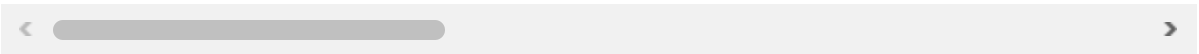


```
In [10]: # sg_edmgc.info()      # check the structure of the data frame
sg_edmgc.head()           # printing the first 5 rows of the data frame
```

Out[10]:

|   | SiteN | State | Lat       | Long       | DO   | pH   | avg_WW   | avg_Depth | pflow_Pool | pflow |
|---|-------|-------|-----------|------------|------|------|----------|-----------|------------|-------|
| 0 | 1     | TX    | 33.741217 | -95.927376 | 10.7 | 6.99 | 5.76666  | 23.75000  | 0.000000   | 83.33 |
| 1 | 2     | TX    | 33.717896 | -95.911323 | 4.2  | 7.22 | 2.94000  | 27.00000  | 60.000000  | 20.00 |
| 2 | 3     | TX    | 33.747466 | -96.019146 | 7.4  | 7.58 | 11.21666 | 46.33333  | 66.666667  | 33.33 |
| 3 | 4     | TX    | 33.554952 | -96.169601 | 7.7  | 7.86 | 10.51666 | 18.41666  | 0.000000   | 66.66 |
| 4 | 5     | TX    | 33.823594 | -95.861067 | 6.8  | 7.42 | 22.00000 | 39.66666  | 100.000000 | 0.00  |

5 rows × 202 columns



# Bar Plots

In this Section I will generate encompassing bar plots to get good overviews of the # of fish species, genus, site attributes and metrics.

I will hope to encompass any relevant but overwhelming data into basic bar plots to get a simple start on understanding this data visually and identify any trends or under-represented features.

## Total Fish Species Bar Plot

This cell simply sums all of the fish species columns in the **seth\_fishCountData\_2025.csv** dataset and visualizes it as a bar plot.

Here I am looking to see the distribution of species so that I can determine species is abundance and few. My hope is that this will be useful in

determining outliers later, and possibly visualize species in fewer numbers than expected.

```
In [ ]: # ===== Data Analysis =====

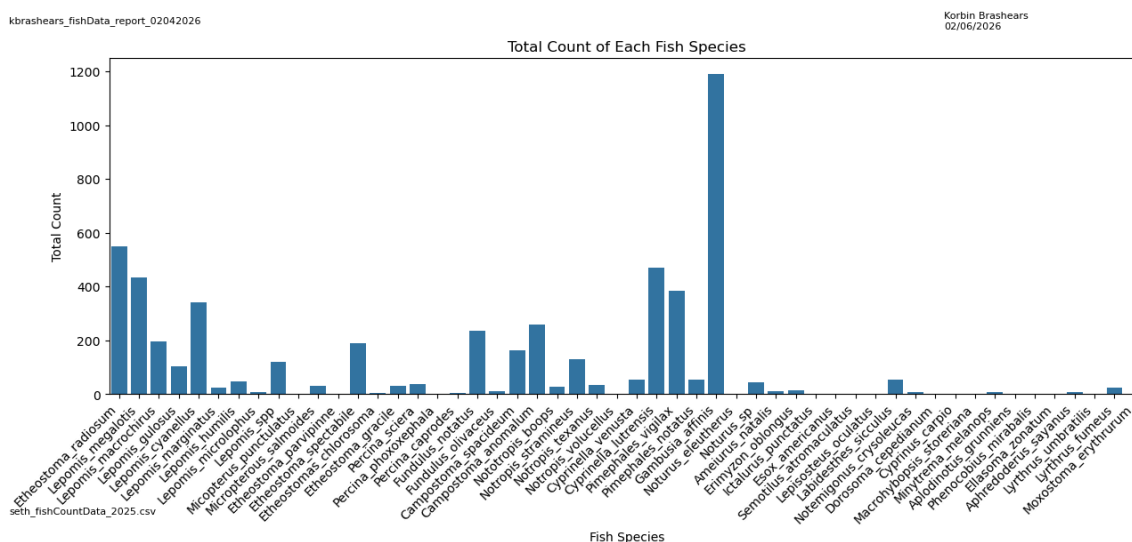
s_fcd_species_cols = s_fcd.columns[2:] # selecting
fish_species_sum = s_fcd[s_fcd_species_cols].sum() # summing e

# ===== Plotting =====

plt.figure(figsize=(15, 6)) # creating
sns.barplot(x=fish_species_sum.index, y=fish_species_sum.values) # creating
plt.xticks(rotation=45, ha='right') # rotating
plt.xlabel('Fish Species') # setting x
plt.ylabel('Total Count') # setting y
plt.title('Total Count of Each Fish Species') # setting t

label_plot_edges(
    program_names="kbrashears_fishData_report_02042026", # calling t
    author_names="Korbin Brashears",
    data_filenames="seth_fishCountData_2025.csv",
    fig_filename="../figures/fish_species_barPlot.png",
    save=False
)

plt.show() # displaying
```



## Total Fish Genus Bar Plot

This cell preforms the same action as before on the **seth\_genusCountData\_2025.csv**, summing the columns and visualizing as a bar plot.

Hopefully this can help to find outliers with unexpected differences between the population in an over all genus vs a species.

```

In [ ]: # ===== Data Analysis =====

s_gcd_genus_cols = s_gcd.columns[2:] # selecting
fish_genus_sum = s_gcd[s_gcd_genus_cols].sum() # summing e

# ===== Plotting =====

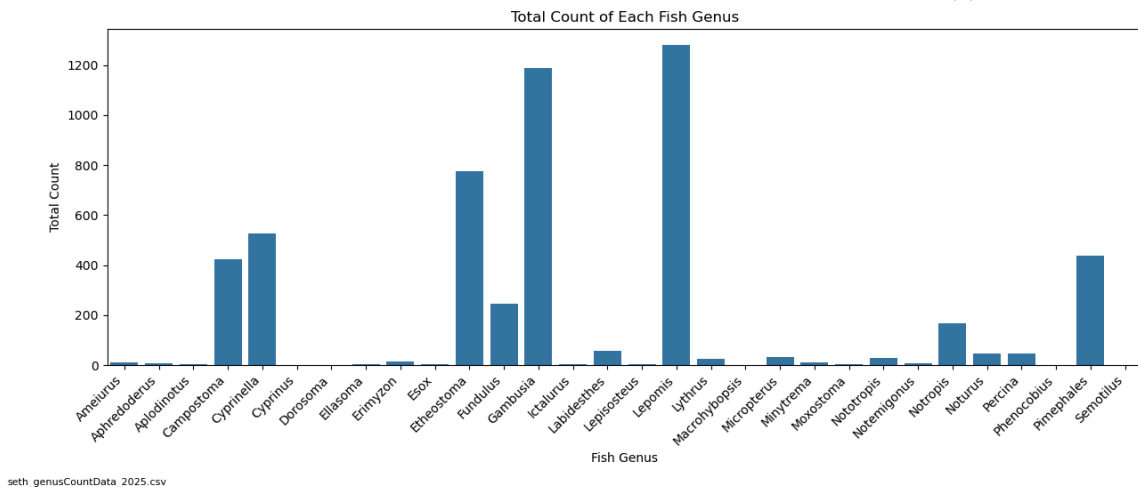
plt.figure(figsize=(15, 6)) # creating
sns.barplot(x=fish_genus_sum.index, y=fish_genus_sum.values) # creating
plt.xticks(rotation=45, ha='right') # rotating
plt.xlabel('Fish Genus') # setting x
plt.ylabel('Total Count') # setting y
plt.title('Total Count of Each Fish Genus') # setting t

label_plot_edges(
    program_names="kbrashears_fishData_report_02042026", # calling t
    author_names="Korbin Brashears",
    data_filenames="seth_genusCountData_2025.csv",
    fig_filename="../figures/fish_genus_barPlot.png",
    save=False
)

plt.show() # displayin

```

kbrashears\_fishData\_report\_02042026

Korbin Brashears  
02/06/2026

seth\_genusCountData\_2025.csv

## Total Population and Genus at Each Site

```

In [ ]: # ===== Plotting =====

plt.figure(figsize=(15, 8)) # creating

plt.subplot(211)
sns.barplot(x=s_egcd['SiteN'], y=s_egcd['t_population']) # creating
plt.xticks(rotation=45, ha='right') # rotating
plt.xlabel('Site Number') # setting x
plt.ylabel('Total Fish Population') # setting y

```

```
plt.title('Total Fish Population at Each Site') # setting t

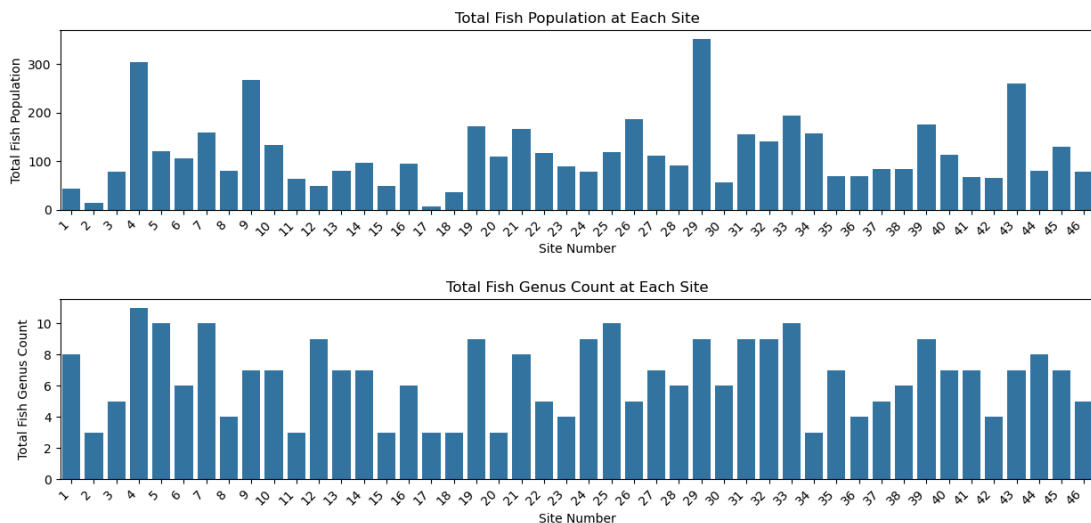
plt.subplot(212)
sns.barplot(x=s_egcd['SiteN'], y=s_egcd['t_genus']) # creating
plt.xticks(rotation=45, ha='right') # rotating
plt.xlabel('Site Number') # setting x
plt.ylabel('Total Fish Genus Count') # setting y
plt.title('Total Fish Genus Count at Each Site') # setting t

plt.subplots_adjust(hspace=0.5) # adjusting

label_plot_edges(
    program_names="kbrashears_fishData_report_02042026", # calling t
    author_names="Korbin Brashears",
    data_filenames="seth_envGenusCountData_2025.csv",
    fig_filename=" ../figures/fish_population_barPlot.png",
    save=False
)

plt.show() # displayin
```

kbrashears\_fishData\_report\_02042026

Korbin Brashears  
02/06/2026

seth\_envGenusCountData\_2025.csv

## Total Population and Genus at Each Site Ordered Bar Plot

```
In [ ]: # ===== Data Analysis =====

s_egcd_tps = s_egcd.sort_values(by='t_population', ascending=False) # sorting t
s_egcd_tgs = s_egcd.sort_values(by='t_genus', ascending=False) # sorting t

# ===== Plotting =====

plt.figure(figsize=(15, 8)) # creating

plt.subplot(211)
```



```

sns.barplot(x=s_egcd_tps['SiteN'], y=s_egcd_tps['t_population'], order=s_egcd_tps['
plt.xticks(rotation=45, ha='right') # rotating
plt.xlabel('Site Number') # setting x
plt.ylabel('Total Fish Population') # setting y
plt.title('Total Fish Population at Each Site') # setting t

plt.subplot(212)
sns.barplot(x=s_egcd_tgs['SiteN'], y=s_egcd_tgs['t_genus'], order=s_egcd_tgs['SiteN
plt.xticks(rotation=45, ha='right') # rotating
plt.xlabel('Site Number') # setting x
plt.ylabel('Total Fish Genus Count') # setting y
plt.title('Total Fish Genus Count at Each Site') # setting t

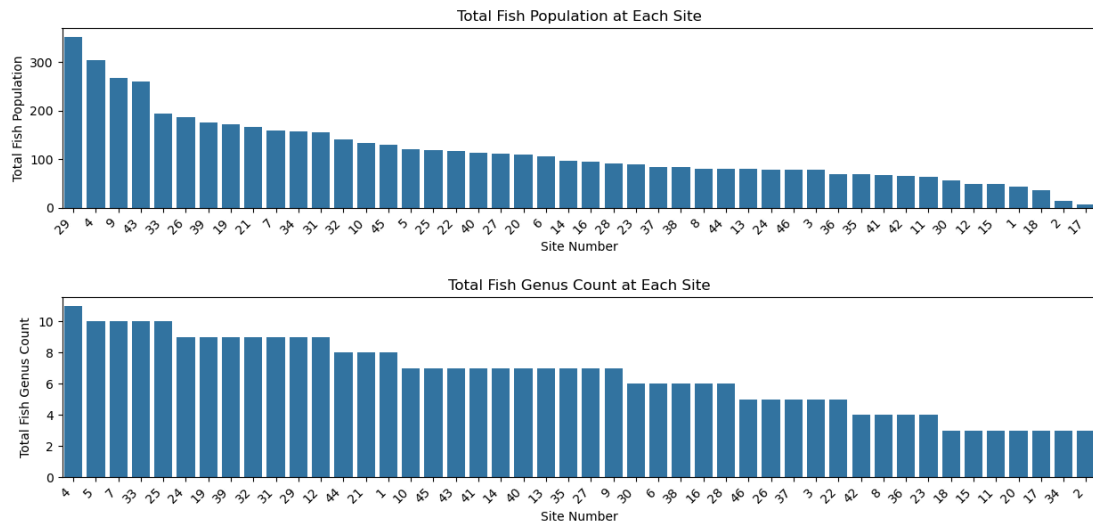
plt.subplots_adjust(hspace=0.5) # adjusting

label_plot_edges(
    program_names="kbrashears_fishData_report_02042026", # calling t
    author_names="Korbin Brashears",
    data_filenames="seth_envGenusCountData_2025.csv",
    fig_filename=" ../figures/fish_population_barPlot.png",
    save=False
)

plt.show() # displayin

```

kbrashears\_fishData\_report\_02042026

Korbin Brashears  
02/06/2026

seth\_envGenusCountData\_2025.csv

## Site Attribute Bar Plots

A function for selecting and summing the attribute columns from the file **seth\_envData\_2025.csv** for a given site N. This will help to see the over all site attribute distribution at a site of interest and manually spot similarities or outliers.

```

In [ ]: # ===== Defining Function =====

def site_attributes_bar_plotter(siteN, save=False):

    # ===== Data Analysis =====

    # Row Selection
    site_att_v = s_ed[siteN-1:siteN]

    att_cols = site_att_v.columns[4:]
    site_att_v = site_att_v[att_cols]

    # Normalization
    max_val = site_att_v.values.max()
    site_att_v = site_att_v / max_val

    # Sorting
    site_att_ov = site_att_v.T.sort_values(by=siteN-1, ascending=False).T

    # ===== Plotting =====

    plt.figure(figsize=(10, 10))

    #-----
    # Subplot 1: Unordered Bar Plot
    #-----

    # Creating Subplot
    plt.subplot(211)

    # Creating Bar Plot
    sns.barplot(x=site_att_v.columns, y=site_att_v.values[0])

    # Setting Lables and Rotation
    plt.xticks(rotation=45, ha='right')
    plt.xlabel('')
    plt.ylabel('Attribute Value')

    # Setting Title
    plt.title(f'Environmental Attributes for Site #{siteN}')

    #-----
    # Subplot 2: Ordered Bar Plot
    #-----

    # Creating Subplot
    plt.subplot(212)

    # Creating Bar Plot
    sns.barplot(x=site_att_ov.columns, y=site_att_ov.values[0])

```

```

# Setting Lables and Rotation
plt.xticks(rotation=45, ha='right')
plt.xlabel('Environmental Attributes')
plt.ylabel('Attribute Value')

# Setting Title
plt.title(f'Ordered Environmental Attributes for Site #{siteN}')

# Adding Vertical Space Between Subplots
plt.subplots_adjust(hspace=0.75)

# ===== Adding Plot Edge Labels =====

label_plot_edges(
    program_names="kbrashears_fishData_report_02042026",
    author_names="Korbin Brashears",
    data_filenames="seth_envData_2025.csv",
    fig_filename=f"../figures/site_{siteN}_attributes_barPlot.png",
    save=save
)

plt.show()

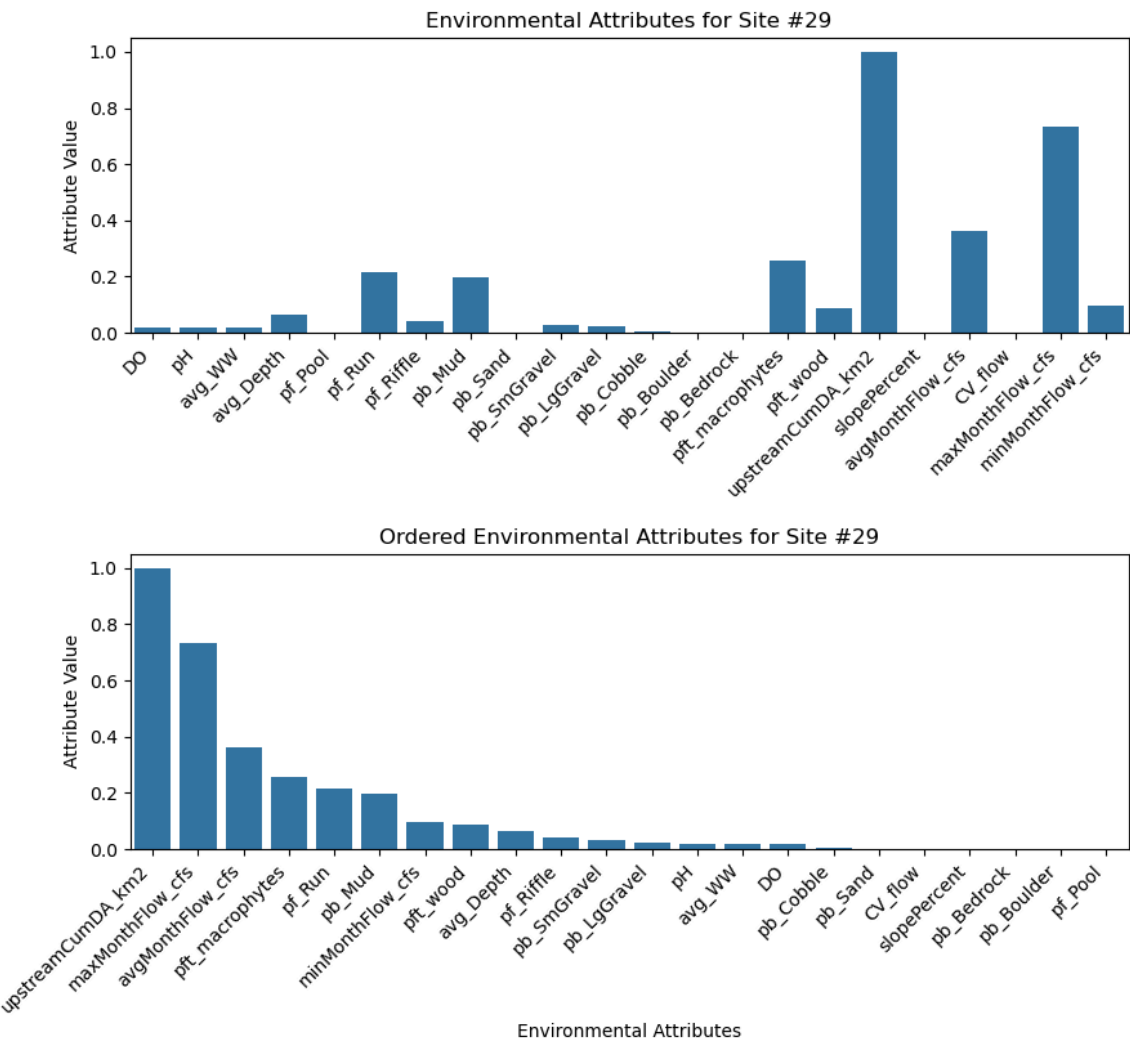
```

## High t<sub>pop</sub> population Sites

Attribute bar plots for the top 4 t<sub>pop</sub> population sites in order (high to low t<sub>pop</sub>)

### Site #29 Bar Plots

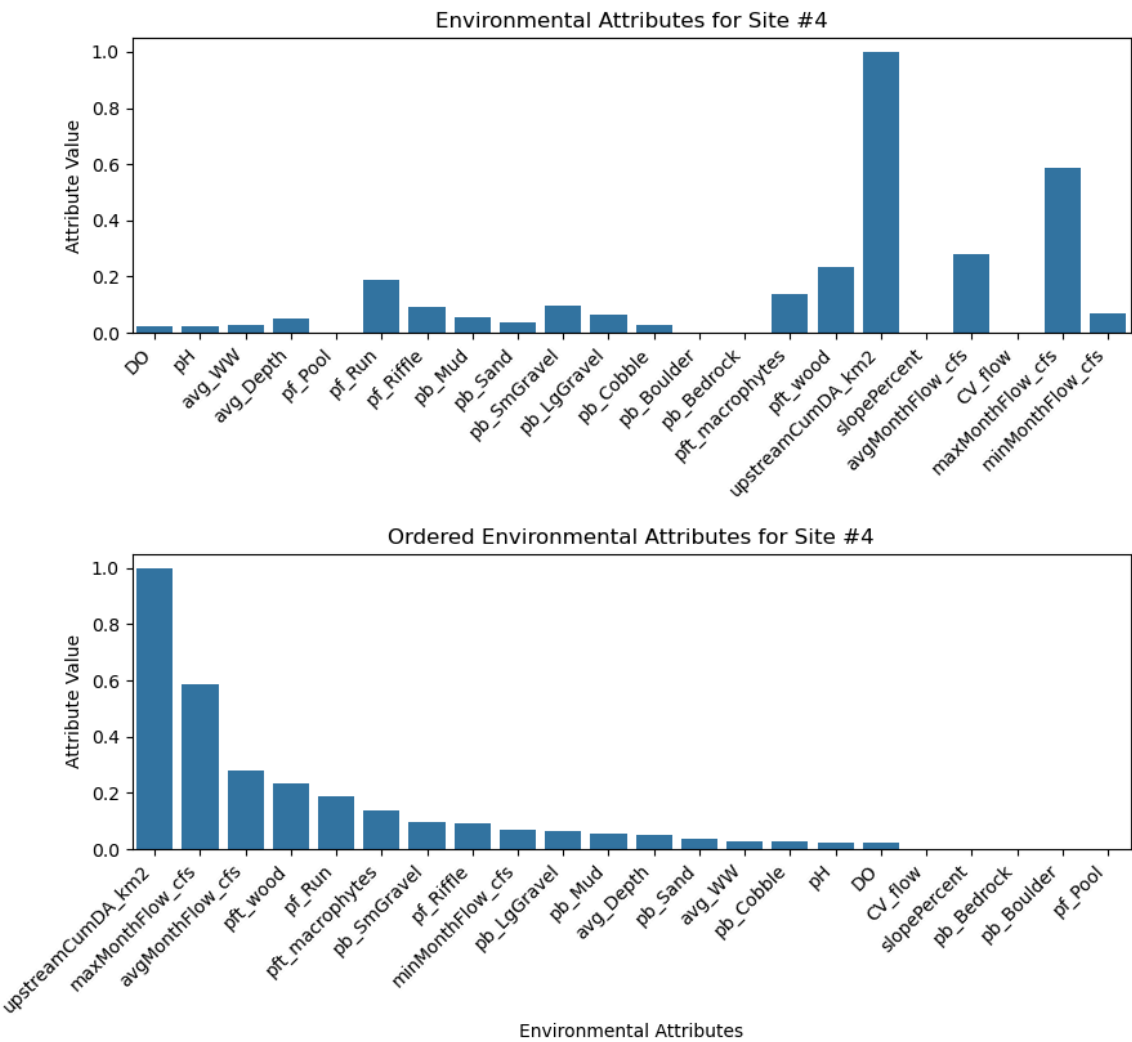
```
In [16]: site_attributes_bar_plotter(siteN=29, save=False) # calling the function to c
```



seth\_envData\_2025.csv

Site #4 Bar Plots

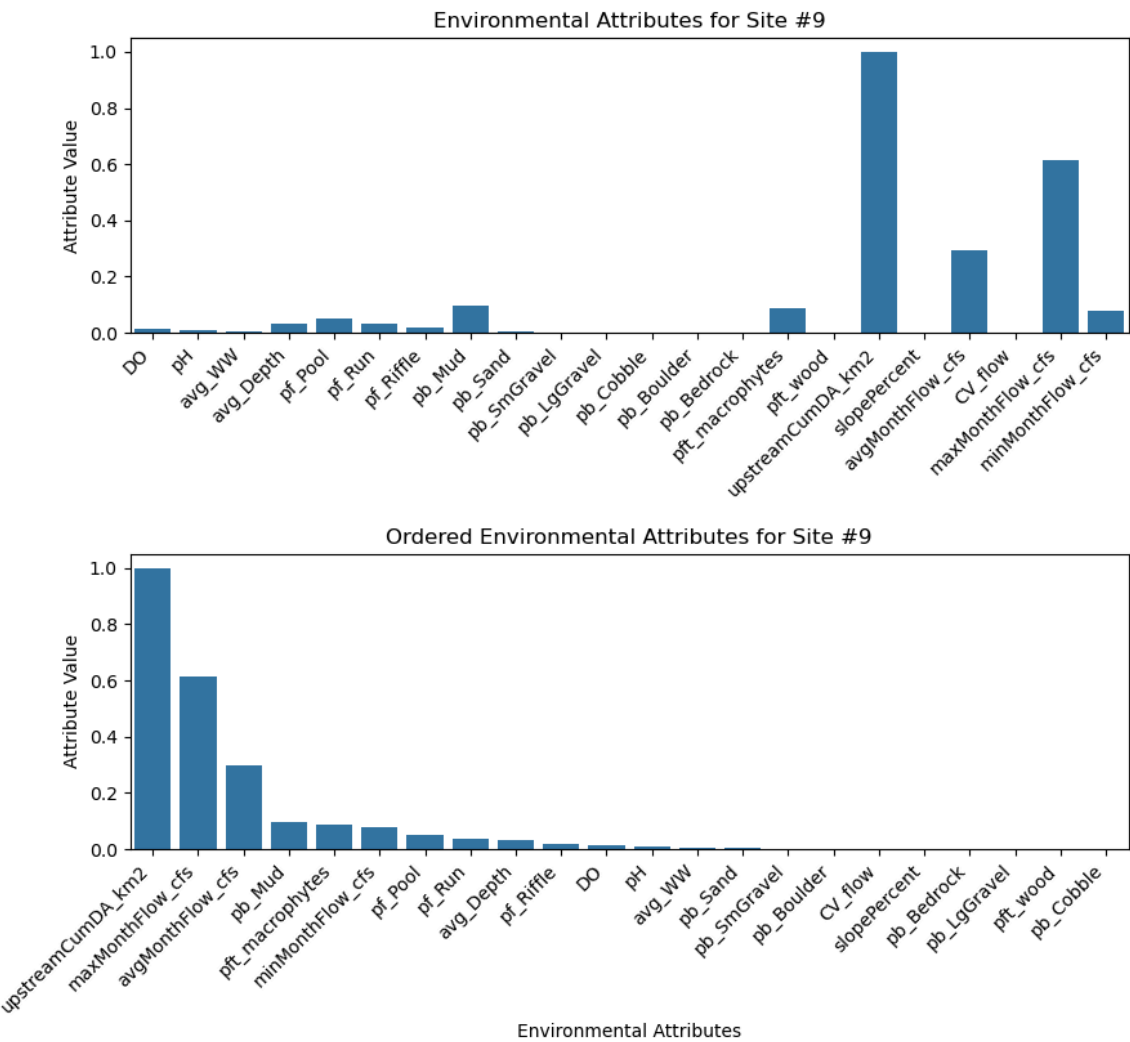
```
In [17]: site_attributes_bar_plotter(siteN=4, save=False) # calling the function to cr
```



seth\_envData\_2025.csv

Site #9 Bar Plots

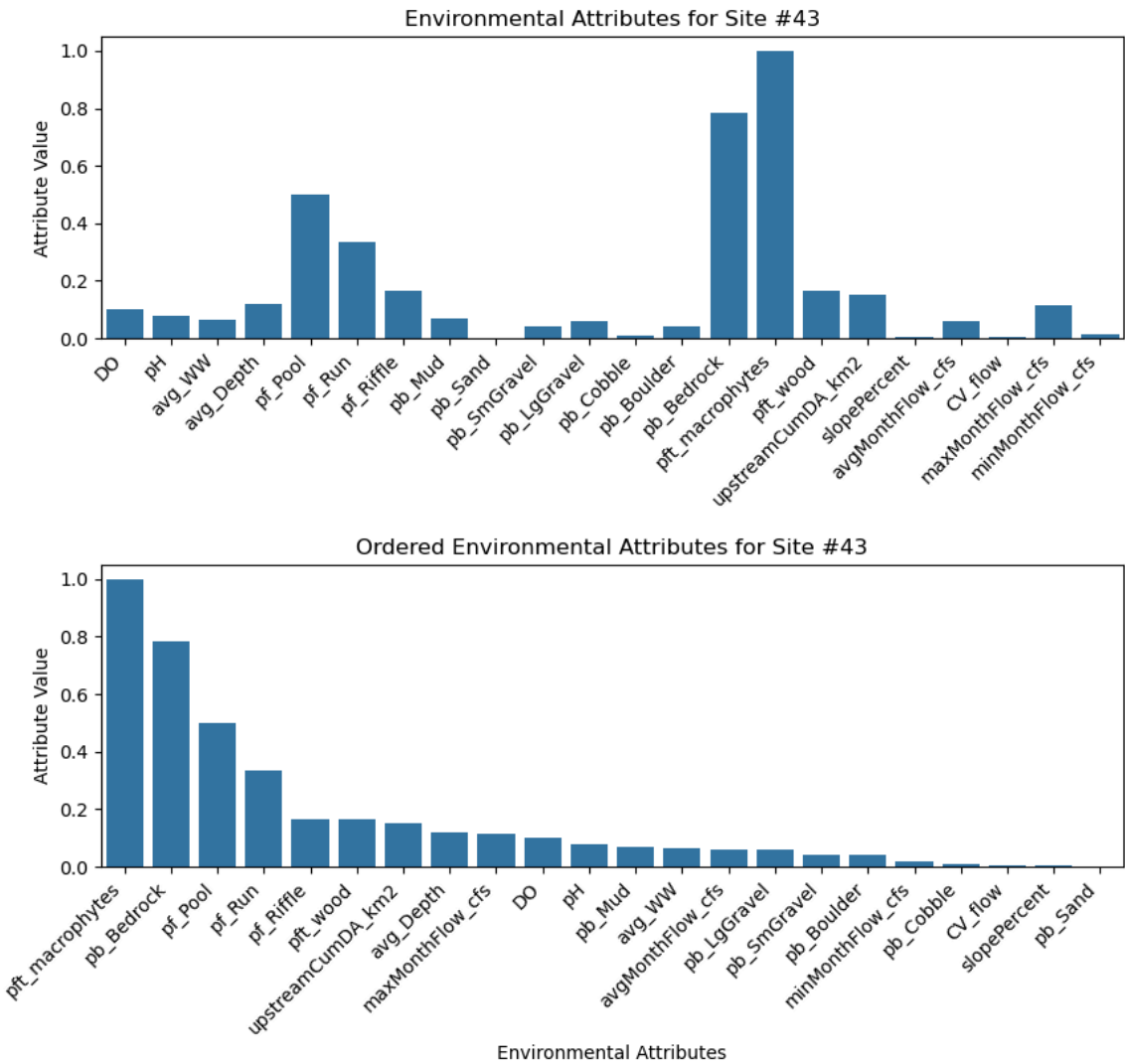
```
In [18]: site_attributes_bar_plotter(siteN=9, save=False) # calling the function to cr
```



seth\_envData\_2025.csv

Site #43 Bar Plots

```
In [19]: site_attributes_bar_plotter(siteN=43, save=False) # calling the function to c
```



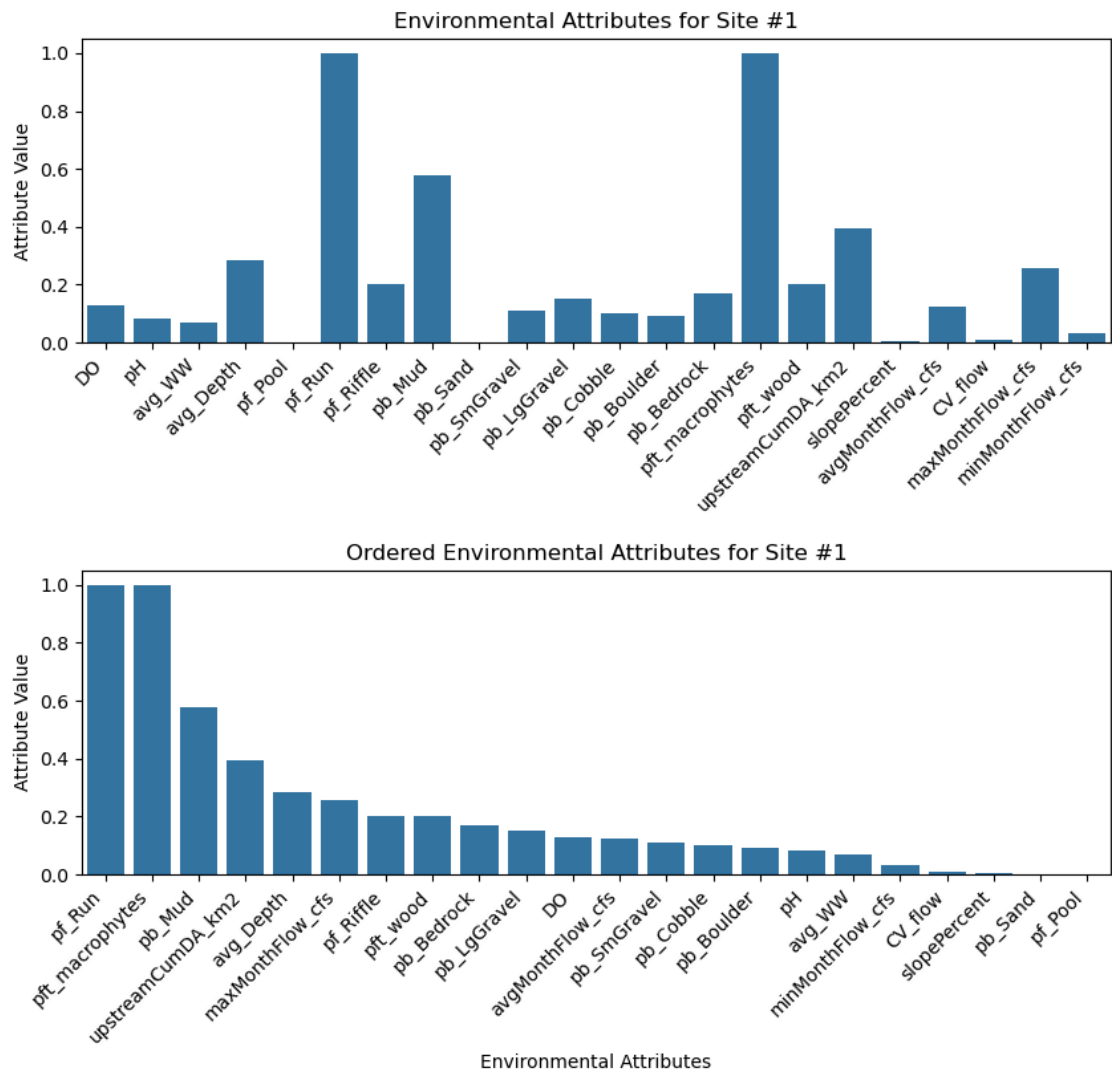
seth\_envData\_2025.csv

Low t\_population Sites

Attribute bar plots for the bottom 4 t\_population sites in order (high to low t\_pop)

Site #1 Bar Plots

```
In [20]: site_attributes_bar_plotter(siteN=1, save=False) # calling the function to cr
```

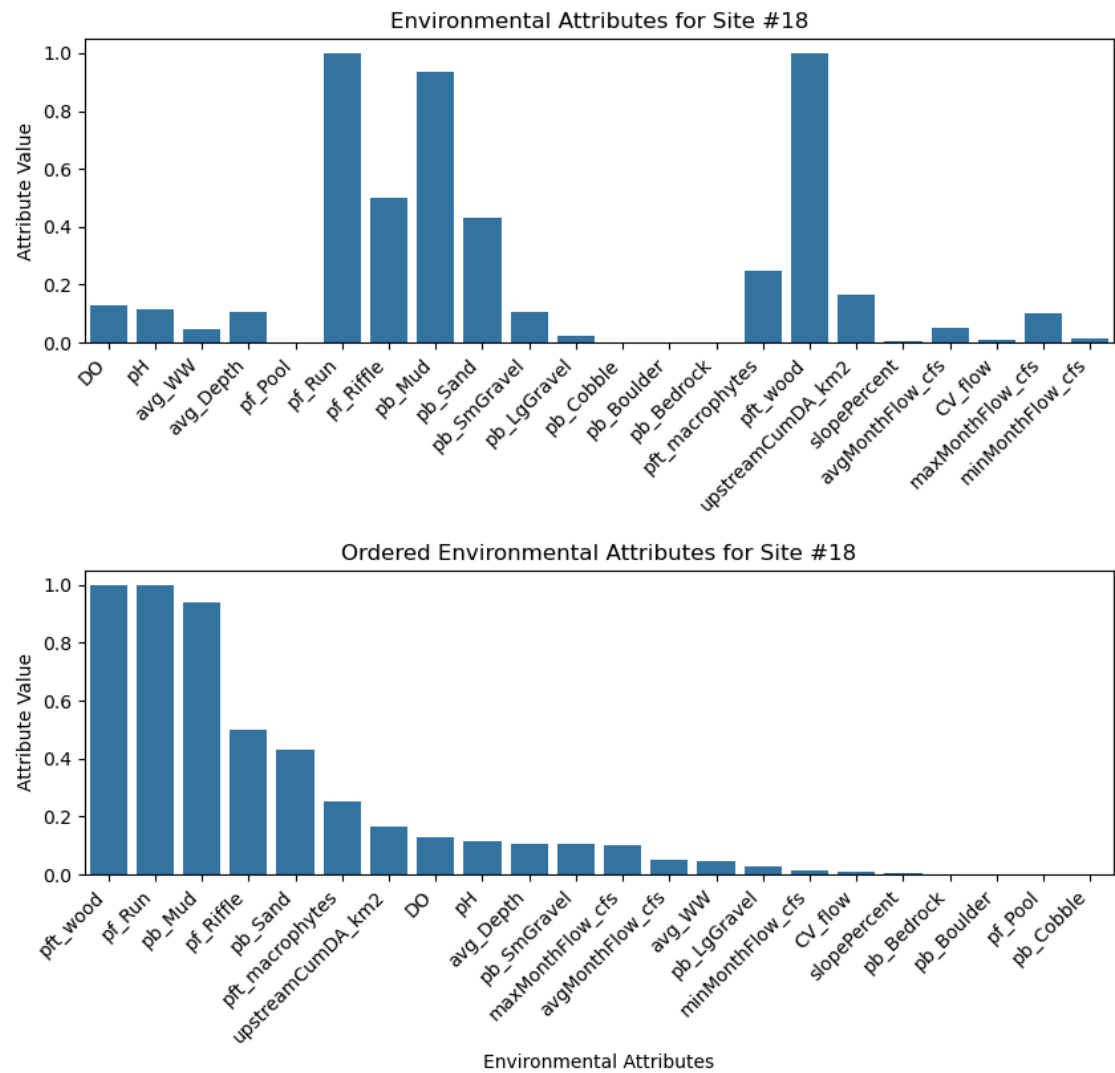


seth\_envData\_2025.csv

Site #18 Bar Plots

```
In [21]: site_attributes_bar_plotter(siteN=18, save=False) # calling the function to c
```



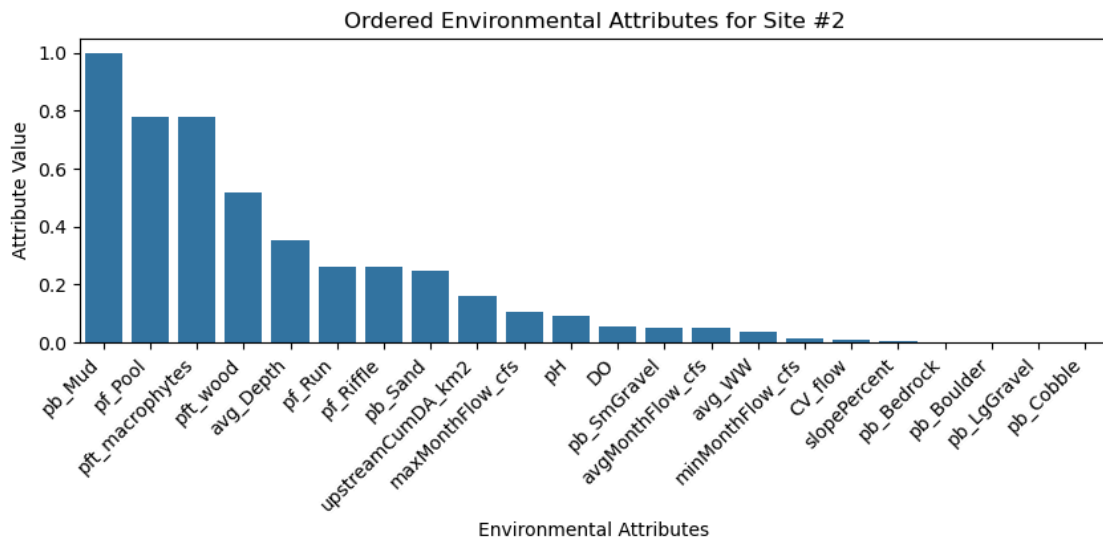
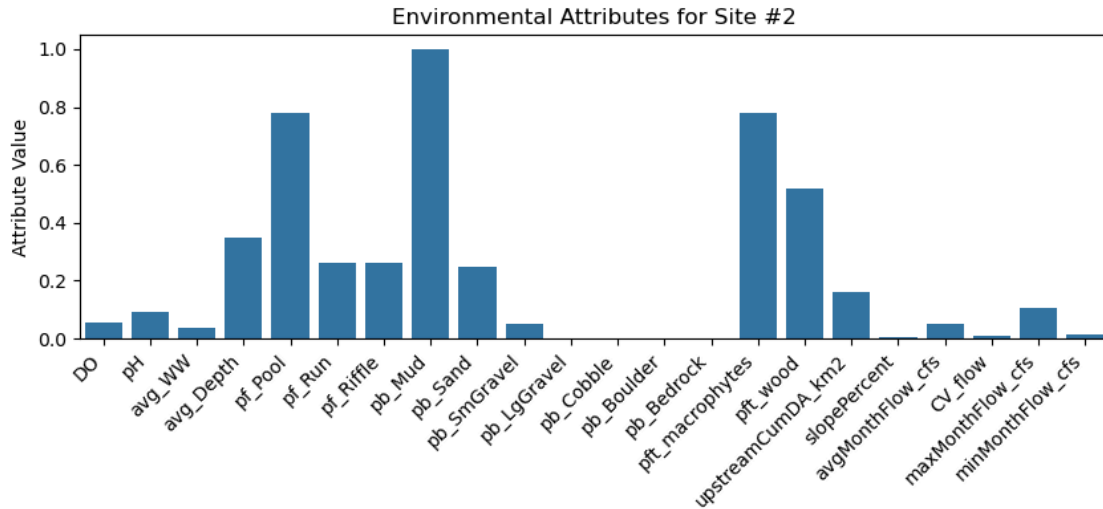


seth\_envData\_2025.csv

Site #2 Bar Plots

```
In [22]: site_attributes_bar_plotter(siteN=2, save=False) # calling the function to cr
```

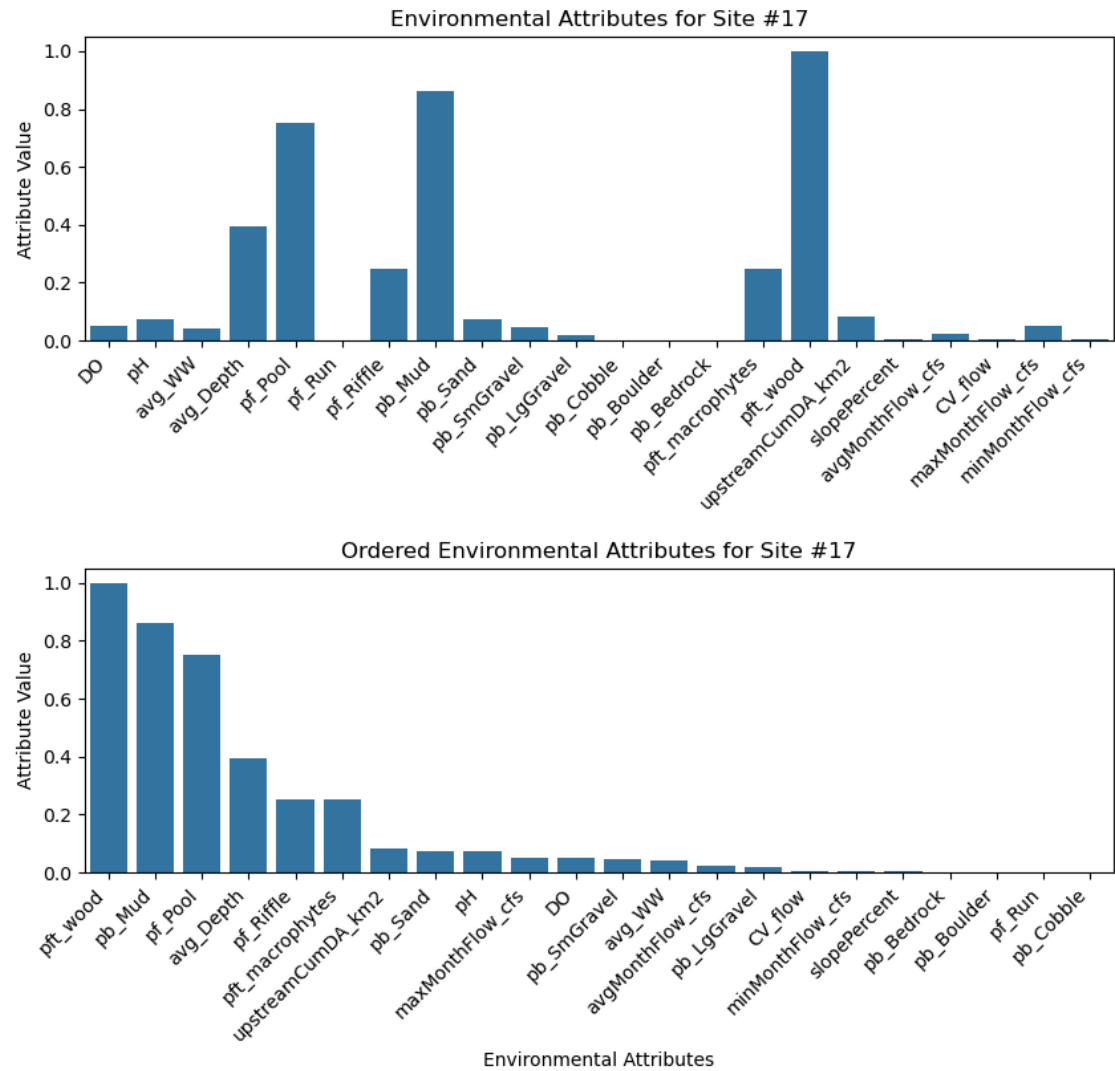
kbrashears\_fishData\_report\_02042026

Korbin Brashears  
02/06/2026

seth\_envData\_2025.csv

### Site #17 Bar Plots

```
In [23]: site_attributes_bar_plotter(siteN=17, save=False) # calling the function to c
```



seth\_envData\_2025.csv